



STEP 6 — SAAS MODE (MULTI-TENANT SYSTEM)

This is where you go from:

“Freelancer” ❌

to

“Product Owner (Monthly Income)” ✅



STEP 6 GOAL

Build a system where:

- Multiple clients (companies) can use your CRM
- Each client has **separate data**
- You (admin) control everything
- You can charge monthly



6.1 ARCHITECTURE OVERVIEW

Super Admin (You)



Company (Client 1)

- Leads
- Funnels
- Conversations
- Appointments

Company (Client 2)

- Separate Data



6.2 CORE CONCEPT: **company_id**

This is the **most important field in entire system**

👉 Every data must belong to a company

✓ Update All Tables

Add `company_id`:

```
ALTER TABLE tblleads ADD company_id INT DEFAULT 0;  
ALTER TABLE tbl_conversations ADD company_id INT DEFAULT 0;  
ALTER TABLE tbl_appointments ADD company_id INT DEFAULT 0;  
ALTER TABLE tbl_funnels ADD company_id INT DEFAULT 0;  
ALTER TABLE tbl_funnel_pages ADD company_id INT DEFAULT 0;
```



6.3 CREATE COMPANIES TABLE

✓ Table: `tbl_companies`

```
CREATE TABLE tbl_companies (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(191),  
  email VARCHAR(191),  
  phone VARCHAR(50),  
  plan_id INT,  
  status VARCHAR(50) DEFAULT 'active',  
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP  
);
```



6.4 USER → COMPANY MAPPING

✓ Modify Staff Table

```
ALTER TABLE tblstaff ADD company_id INT DEFAULT 0;
```



Logic:

- Each staff belongs to ONE company

- Super admin → company_id = 0
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6.5 DATA ISOLATION (CRITICAL)



GLOBAL RULE:

👉 Every query MUST filter by `company_id`



WRONG:

```
$this->db->get('tblleads')->result();
```



CORRECT:

```
$this->db->where('company_id', get_company_id());  
$this->db->get('tblleads')->result();
```



Helper Function:

```
function get_company_id()  
{  
    return $_SESSION['company_id'] ?? 0;  
}
```



6.6 AUTO-INJECT COMPANY ID



On Insert:

```
$data['company_id'] = get_company_id();  
$this->db->insert('tblleads', $data);
```



6.7 SUBDOMAIN SYSTEM (LIKE GHL)



Example:

client1.yourcrm.com
client2.yourcrm.com



Approach:

1. Wildcard domain:

*.yourcrm.com

2. Detect subdomain:

```
$host = $_SERVER['HTTP_HOST'];  
$subdomain = explode('.', $host)[0];
```

3. Map to company:

```
$company = $this->db->where('subdomain', $subdomain)->get('tbl_companies')->row();
```



Add Column:

```
ALTER TABLE tbl_companies ADD subdomain VARCHAR(191);
```



6.8 SUPER ADMIN PANEL



URL:

/super-admin

Features:

- Create company
- Assign plan

- View all clients
 - Disable account
 - View usage
-
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6.9 SUBSCRIPTION SYSTEM (BASIC)

✓ Table: **tbl_plans**

```
CREATE TABLE tbl_plans (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(100),  
  price DECIMAL(10,2),  
  max_users INT,  
  max_contacts INT,  
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP  
);
```

✓ Assign Plan:

tbl_companies.plan_id



Enforce Limits:

Example:

```
if ($total_leads > $plan->max_contacts) {  
  show_error('Limit reached');  
}
```



6.10 SECURITY LAYER

✓ Must Implement:

- Prevent cross-company access

- Validate every request
 - Use middleware/hook
-

Example:

```
if ($lead->company_id != get_company_id()) {  
    show_404();  
}
```



6.11 FILE STORAGE ISOLATION

Structure:

```
/uploads/company_1/  
/uploads/company_2/
```



6.12 AUTOMATION MULTI-TENANT SUPPORT

Modify Trigger:

```
trigger_event('lead_created', [  
    'company_id' => get_company_id(),  
    'lead_id' => $lead_id  
]);
```

👉 n8n should respect company_id



6.13 TESTING CHECKLIST

Must Test:

- Company created
 - Users assigned
 - Data isolated
 - Subdomain working
 - Plan limits working
 - Super admin control works
 - No data leakage
-



STEP 6 OUTPUT

After this:

-  Multi-tenant SaaS system
 -  Company-based data separation
 -  Subdomain-based access
 -  Subscription-ready structure
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